Digital Image Processing Assignment 03 – Report Analysis

Name : Naila Syakira
NIM : 25/555286/PA/23270

In this experiment, a segmentation and morphological processing pipeline was built to isolate a target object (a silver coin) from its background and clean up the resulting mask. The goal was to understand how region growing behaves on real, textured images, and how fundamental morphological operations—dilation and erosion—can be used to fix the inevitable imperfections left by the segmentation step.

The method used was straightforward. First, a region growing algorithm was implemented from scratch to segment the coin based on a seed pixel and an intensity threshold. Then, binary dilation and erosion were implemented from scratch using a 3×3 structuring element. Finally, a morphological closing operation (repeated dilation followed by repeated erosion) was applied to the raw mask. The results were compared side-by-side to evaluate how morphology transforms the segmentation output.

The final notebook contains the core functions:

1. region_growing takes an image, a seed coordinate, and a threshold. It uses a stack-based flood-fill approach, checking the four immediate neighbors of every pixel. If a neighbor's grayscale intensity is within the threshold of the seed's intensity, it is added to the mask.
2. dilate and erode are the fundamental morphological operators, implemented using nested loops and a hardcoded 3×3 structuring element. Dilation turns a background pixel into a foreground pixel if any of its neighbors are foreground (growing the region). Erosion turns a foreground pixel into background if any of its neighbors are background (shrinking the region).
3. closing_large is a repeated morphological closing operation. Instead of a single 3×3 pass, it applies dilation by N times, followed by erosion N times. This simulates a much larger structuring element, which is necessary to close the large gaps in the coin's design.

1. Region Growing Segmentation

The coin image was selected because it presents a classic real-world segmentation challenge. The silver coin is bright, but it contains intricate embossed details (the Britannia figure, text, and background patterns) that create sharp intensity changes within the object itself.

After testing several values, the seed point was placed at (332, 332) (the center of the coin) with a threshold of T = 37. This resulted in a raw mask covering 31.1% of the image.

Visually, the raw region growing mask successfully captured the general circular outline of the coin. However, it suffered from two major issues:

- First, the mask was incredibly "noisy," with small fragmented patches scattered around the edges.
- Second, and more importantly, the embossed details of the coin caused massive internal gaps. Because the intensity of the raised details differed from the flat metal by more than the threshold, the algorithm treated these areas as background, leaving the coin looking like a hollow, fragmented ring rather than a solid object.



2. Dilation and Erosion

Before combining them, the base operators were tested individually to understand their effects on the raw mask.

Dilation expands the foreground pixels. As expected, applying dilation caused the coin's outer boundary to grow thicker, and the internal gaps began to shrink. However, it also exaggerated the noise, making the fragmented patches outside the coin larger and more prominent. It was clear that dilation alone was not a solution, as it made the mask "dirtier" overall.

Erosion shrinks the foreground pixels. When applied to the raw mask, the coin's boundary was eaten away. This was particularly destructive to the thin, fragmented rings that made up the coin's edges. While erosion is excellent for removing small noise patches (which simply disappear), it aggressively widened the internal holes of the coin, making the segmentation look even worse.

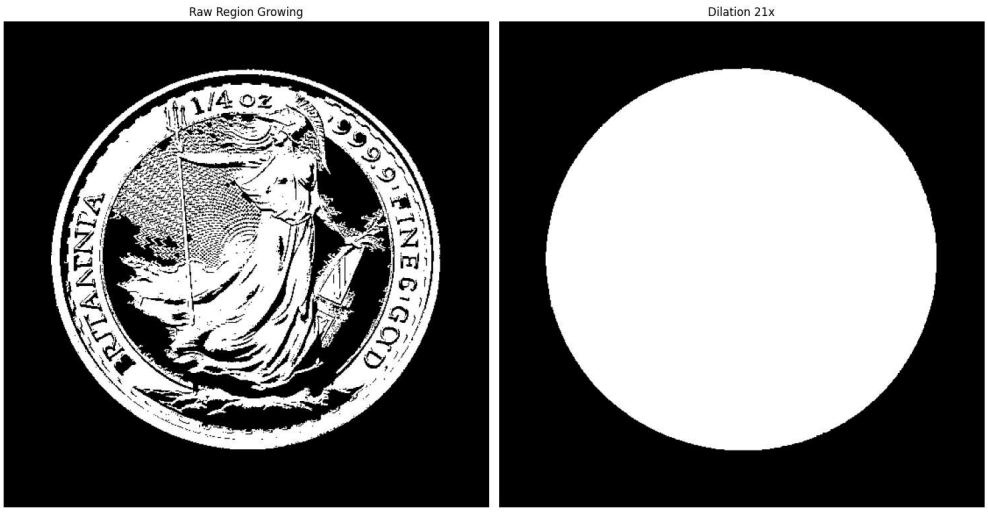


3. Closing with Repeated Dilation and Erosion

To fix the internal gaps without permanently shrinking or destroying the coin's shape, morphological closing was applied. A standard single-pass closing (dilate once, erode once) was insufficient because the embossed gaps were much wider than a 3-pixel bridge.

Instead, a repeated closing was performed. The code applied dilation 21 times (n=21), followed immediately by erosion 21 times. The visual difference is dramatic. The 21x dilation completely overwhelmed the internal gaps, turning the fragmented coin into a solid, heavy white disk, albeit much larger than the original. The subsequent 21x erosion then ate away that excess boundary, shrinking the disk back down to roughly its original size.

The final result successfully filled the internal gaps and smoothed the outer boundary, transforming the noisy, fragmented raw mask into a solid, clean silhouette of the coin. However, this aggressive approach comes with a trade-off: the repeated operations have slightly rounded off the sharp, crisp edges of the coin, and the exact original circular profile has been smoothed into a slightly softer shape. But it has one downside, high value of N will result in longer execution time. In the case of this experiment, it took around ~29s.



Conclusion

Comparing the segmentation result with and without morphological processing highlights the true purpose of morphology in image analysis: it is a cleaning tool, not a segmentation algorithm. Region growing found the general location and extent of the object, but failed on the details. Morphological closing acted as a powerful filter, bridging the internal gaps and suppressing external noise that the thresholding step could not handle.

The experiment also proves that parameter tuning (the number of repeated operations) is critical. A single 3×3 operation was useless against the large gaps of the coin, but an aggressive 21x repetition successfully produced a clean, solid mask, trading a slight loss in boundary precision for a massive gain in overall structural integrity but with a cost of performance.